



NEA

AQA A-LEVEL COMPUTER SCIENCE 2027

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Analysis

Anything written in red are notes for me while writing and will not be present in the final version
Please ignore them while marking.

Background Description

Background of the problem in sufficient detail for a third party to understand the problem

[Blood on the Clocktower](#) is a popular, Mafia-style party game created by [The Pandemonium Institute](#) (TPI), where an informed minority of players need to survive and eliminate the uninformed majority in order to win. However, unlike games such as Werewolf, each player has a unique power.

During gameplay, there is one person, the Storyteller, who is the person running the game, similar to a Game Master in TTRPGs. They have everyone's characters in a "grimoire" which allows them to see which players have what role, mark who's dead etc.

The characters that the players could be come from a "script", essentially a long list of characters. An example of one is Trouble Brewing, the beginner script, pictured in [Appendix 1.A](#). These scripts can consist of both official (released by TPI) as well as homebrew / custom characters.

For more information, TPI's Official YouTube channel has a [How to Play](#) video.

Introduction

An introduction to the problem or investigation

For my NEA, I am creating a virtual grimoire for Blood on the Clocktower. The application will allow the user to assign roles to players, represented by circles and nametags, add "reminder tokens" as well as help them run the game by providing prompts of what to do or what to tell the player.

It will be a potential replacement for the physical grimoire that comes with the game, pictured in [Appendix 1.B](#).

This application will be web-based and created using the [Godot Game Engine](#) in [GDScript](#) (a high-level language syntactically similar to python) as it is a very versatile game engine and can export to my desired platform.

Problem Outline

When playing in person, it can be difficult to transport all tokens to where the game is being played in a secure way and keep them in the grimoire while moving around during the night. Additionally, if the storyteller wishes to use lots of characters, changing them over can be difficult.

A virtual grimoire would eliminate these problems as it drastically reduces the physical items needed, in some cases down to just a laptop or tablet. However, lots of existing virtual grimoires don't allow for the same freedoms as physical grimoires do.

Additionally, virtual grimoire would have the ability to provide help to beginners, allowing them to become more confident in storytelling by assisting them in what they need to do.

Research

Provide research into the current problem that identifies other applications and comparisons to the current problem you are solving

- + screenshots
- + what makes them good
- + what could be improved

clocktower.online

clocktower.online has one of the more complex workflows due to its main purpose of online usage. Moreover, in order to run a game, one must first host a game, then select a script and hand out characters, same as others.

Advantages

- Works well
- Has capacity for random character assignment
- Has support for both custom and homebrew scripts

Disadvantages

- Designed for playing entirely online, not in-person
- It is no longer updated (being replaced by botc.app) so doesn't have newer characters
- Doesn't allow for the free movement of tokens

clocktower.online viewed in Firefox 140.8.0esr (64-bit)



grimmy.app

grimmy.app is likely the app that I will be drawing the most inspiration from. It has a very simple and uncluttered UI while still providing a good user experience. Additionally, it has the option for players to find out their characters without the need for the storyteller to inform them of it.

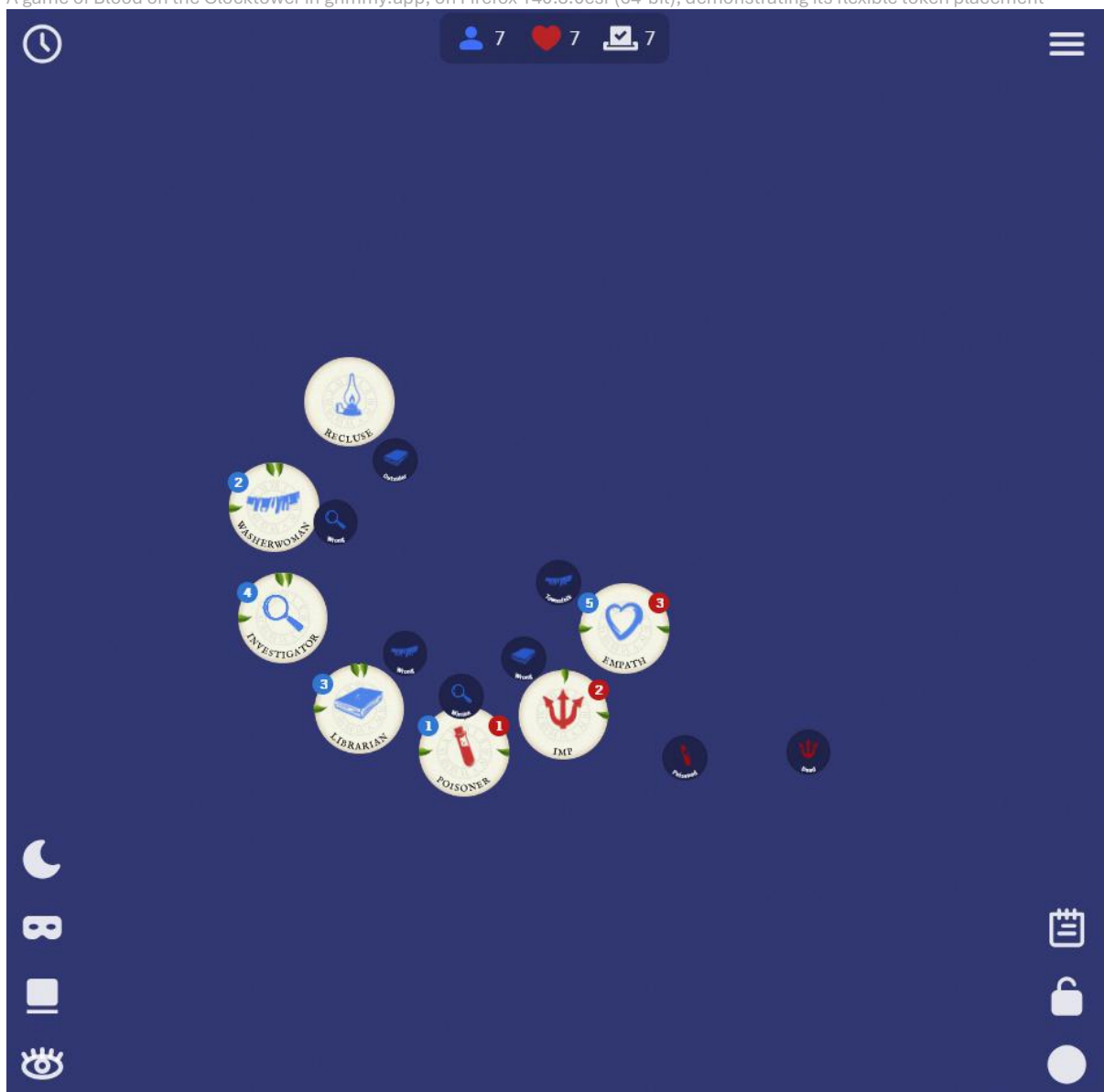
Advantages

- Tokens can be moved around freely
- Has support for custom scripts
- Simulates an actual grimoire very well
- Has night-time information cards

Disadvantages

- Does not have support for homebrew scripts / characters

A game of Blood on the Clocktower in grimmy.app, on Firefox 140.8.0esr (64-bit), demonstrating its flexible token placement



Pocket Grimoire

Pocket Grimoire is also a good example of something similar to what I am trying to make due to its freedom of token movement as well as nametags. However, its UI is not the best and all the separate tabs can be confusing.

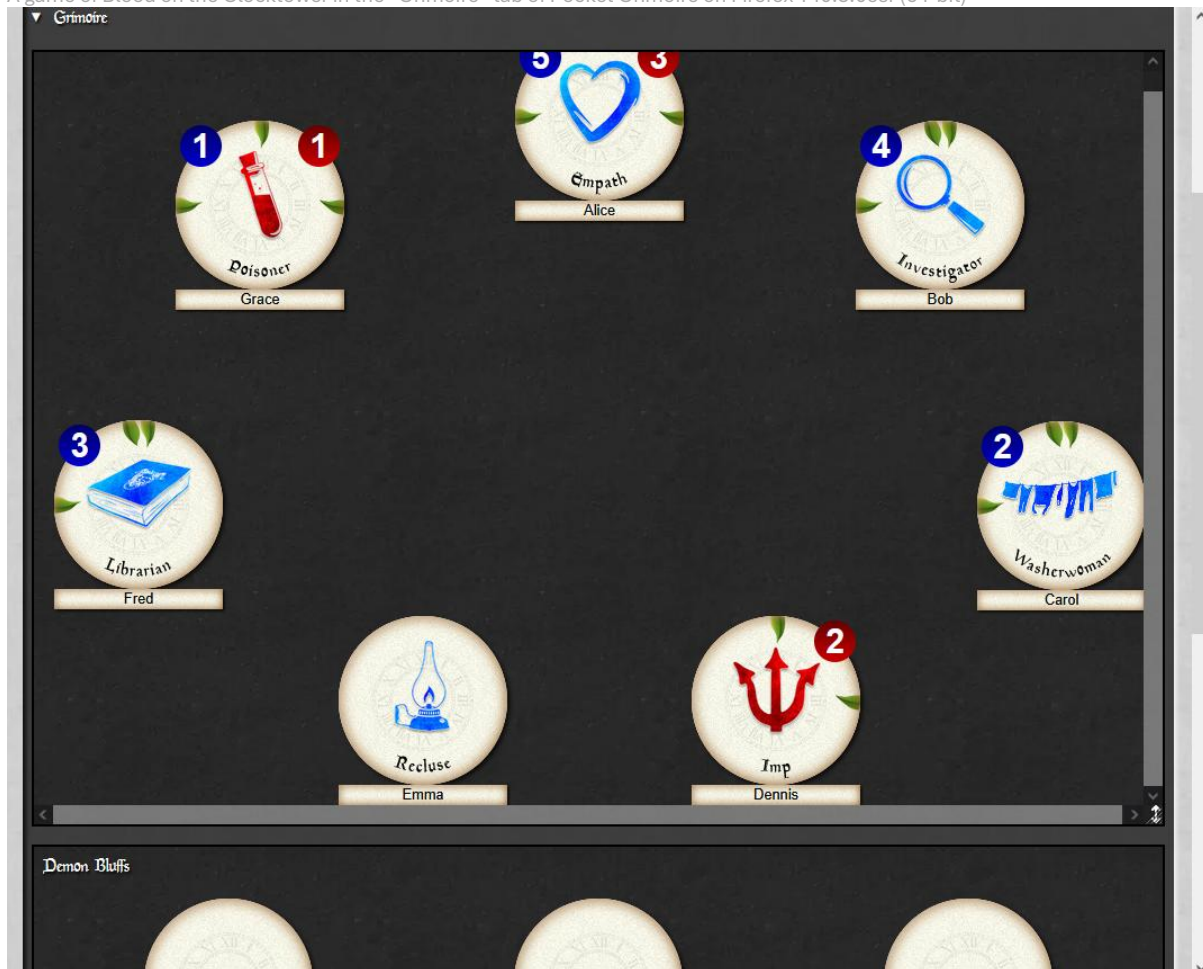
Advantages

- Has capacity for both custom and homebrew scripts
- Simulates the in-person experience very accurately
- Has night-time information cards

Disadvantages

- Not very user-friendly

A game of Blood on the Clocktower in the "Grimoire" tab of Pocket Grimoire on Firefox 140.8.0esr (64-bit)



the-grim.netlify.app

the-grim.netlify.app is a good app for players taking notes however it is not nearly as useful for storytelling. The restrictive token layout is also not useful and while the name tags under characters are good features, the way reminder tokens are displayed is not helpful.

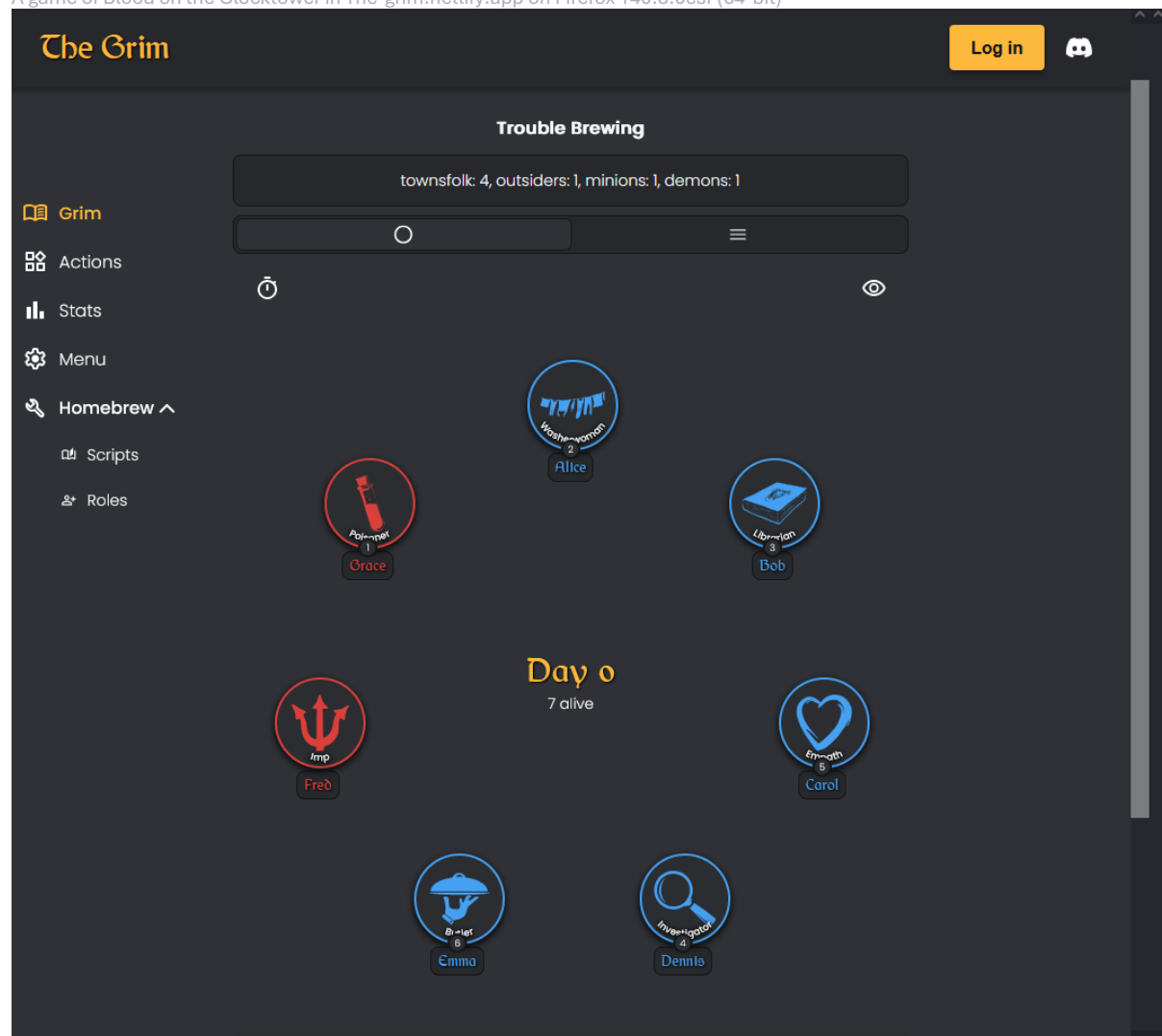
Advantages

- Very easy to use
- Has helpful steps in guiding the setup of a game
- Has some help for beginner storytellers

Disadvantages

- Doesn't allow for the free movement of tokens

A game of Blood on the Clocktower in The-grim.netlify.app on Firefox 140.8.0esr (64-bit)



botc.app

botc.app is the go-to app for online play and a direct improvement on clocktower.online. However, its local game functionality is not as good as some of the others' due to it not being a focus during its creation.

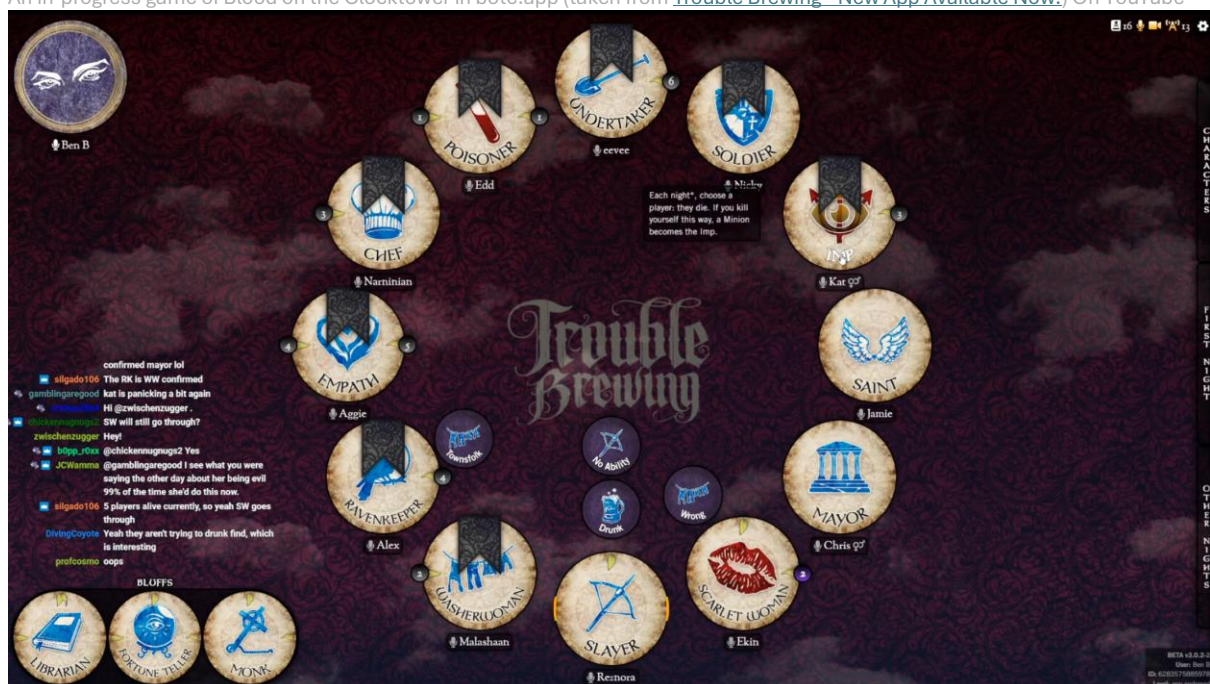
Advantages

- Officially released by The Pandemonium Institute (TPI)
- Is as up-to-date as it is possible to be
- Has support for both custom and homebrew scripts

Disadvantages

- Primarily designed for online play
- Doesn't allow for the free movement of tokens
- Is a paid software (using a subscription)

An in-progress game of Blood on the Clocktower in botc.app (taken from [Trouble Brewing - New App Available Now!](https://www.youtube.com/watch?v=...)) On YouTube



Summary

From the examples I've looked at of similar projects that already exist, my main dislike with them is their lack of flexibility when compared to the physical version of the game.

More specifically, the majority of the sites examined do not have the feature of moveable tokens, they are fixed in a particular shape (commonly a circle) and while this is a common layout, physical space constraints may lead to players being sat in others, such as a square or horseshoe. This is disadvantageous as it is better to be able to move the players' tokens to fit the environment that the game is being played in.

Interview

In order to create the objectives and success criteria for this project, I will be interviewing my friend, PJ, who plays and storytells Blood on the Clocktower, to see what features he would want in an application such as this one.

The transcript of the full interview can be found in [Appendix 3](#).

Questions

The following entries are the summarised responses when PJ was asked the corresponding questions.

1. What assistance would be useful in storytelling (i.e. night order, what to do when waking people)?
 - 1.1. [Night order and character. Also being able to use multiple of the same reminder tokens¹](#).
2. Would alternate colour schemes (such as light/dark mode) be good features?
 - 2.1. [Yes](#)
3. Should the app have different modes, i.e. strict vs flexible²?
 - 3.1. [Yes](#)
4. Would automatic character assignment be a good feature?
 - 4.1. [Yes, as you don't need the physical tokens](#)
5. How closely should the app replicate in-person usage?
 - 5.1. [As closely as possible](#)
6. Would physical or digital night information cards be preferable?
 - 6.1. [Digital](#)
7. Would having capacity for homebrew characters be useful?
 - 7.1. [Yes](#)
8. Would you rather an in-app tutorial or a document-based guide?
 - 8.1. [In-app](#)
9. Should there be a feature that tells you what to tell the player³?
 - 9.1. [Yes, but it should be optional](#)
10. Any other suggestions or ideas for this app?
 - 10.1. [Placeable name tags](#)

Summary

Based off of the interview, the vast majority of the suggested features were seen to be wanted. Furthermore, the addition of name tags for characters will drastically improve the storytelling experience, especially considering they are not a part of the base game.

¹ Certain character's abilities (such as the [Cannibal](#)) use other characters abilities so may need additional copies of those characters' tokens

² This question is in reference to freedom of token movement, see [grimmy.app](#) (flexible) and [the-grim.netlify.app](#) (strict)

³ For example, telling the storyteller how many of the [Empath](#)'s neighbours are evil

Objectives

Numbered list of measurable specific objectives covering all required functionality of the solution or areas of investigation

1. Basic Functionality
 - 1.1. Character tokens can be added to or removed from a grimoire
 - 1.2. Character tokens can be changed without needing to add a new one
 - 1.3. Reminder tokens can be added to or removed from a grimoire
 - 1.4. Character tokens can be initially arranged neatly into a set shape
 - 1.5. Have a section for night information (similar to [botc.app](#)'s waking system)
2. Additional Functionality
 - 2.1. Allow players to learn their character through the app (according to their seat)
 - 2.2. Option to switch between Strict and Flexible modes
 - 2.3. Placeable name tags
3. Basic Storyteller Assistance
 - 3.1. Include a toggleable night order sheet
 - 3.2. Include night order indicators on the character tokens
 - 3.3. Have character abilities show on click / hover of a token
4. Additional Storyteller Assistance
 - 4.1. Have a walk-through for what the storyteller needs to do / tell the player
5. Tutorial
 - 5.1. Include a basic tutorial for how to setup and run a game
 - 5.2. Create a docs page for more detailed instructions and homebrew tutorials

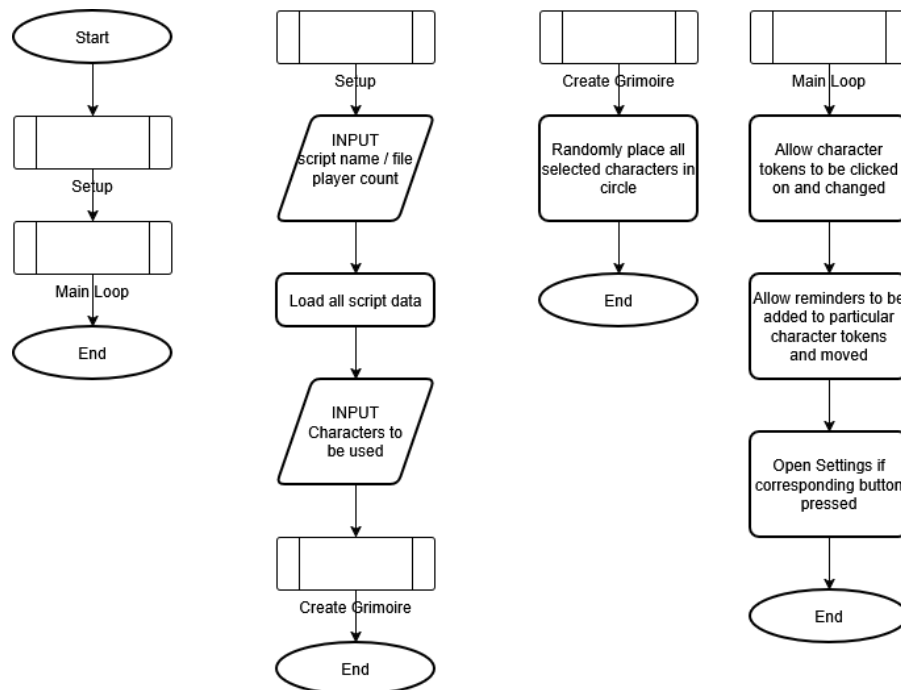
Success Criteria

In order for this project to be useable, it needs to have completed objectives 1.1-1.3 at a bare minimum and having objective 5.1 completed would be extremely useful for any potential new users, especially ones who are new to Blood on the Clocktower.

Modelling of the Problem

Any modelling that will inform the design stage, graph / network model or an entity relation model, scientific / mathematical models / formulae, data flow diagrams

The below flow chart (I have no idea what's going on with the subroutine labels) shows roughly how many existing virtual grimoire programs work.



Design

Introduce what a design is.

ANNOTATED flowcharts

Graph schemas

Entity relation diagram

UI Design

Mock-ups (probably PowerPoint) + annotations i.e. colour codes

Pre-Test plans

Test Type	Object being Tested	Actual Test	Expected Outcome
Login	Login Button	Username	Username should be recognised

Reference to unit circle to get in a circle

Reference to custom character actions yaml stuff

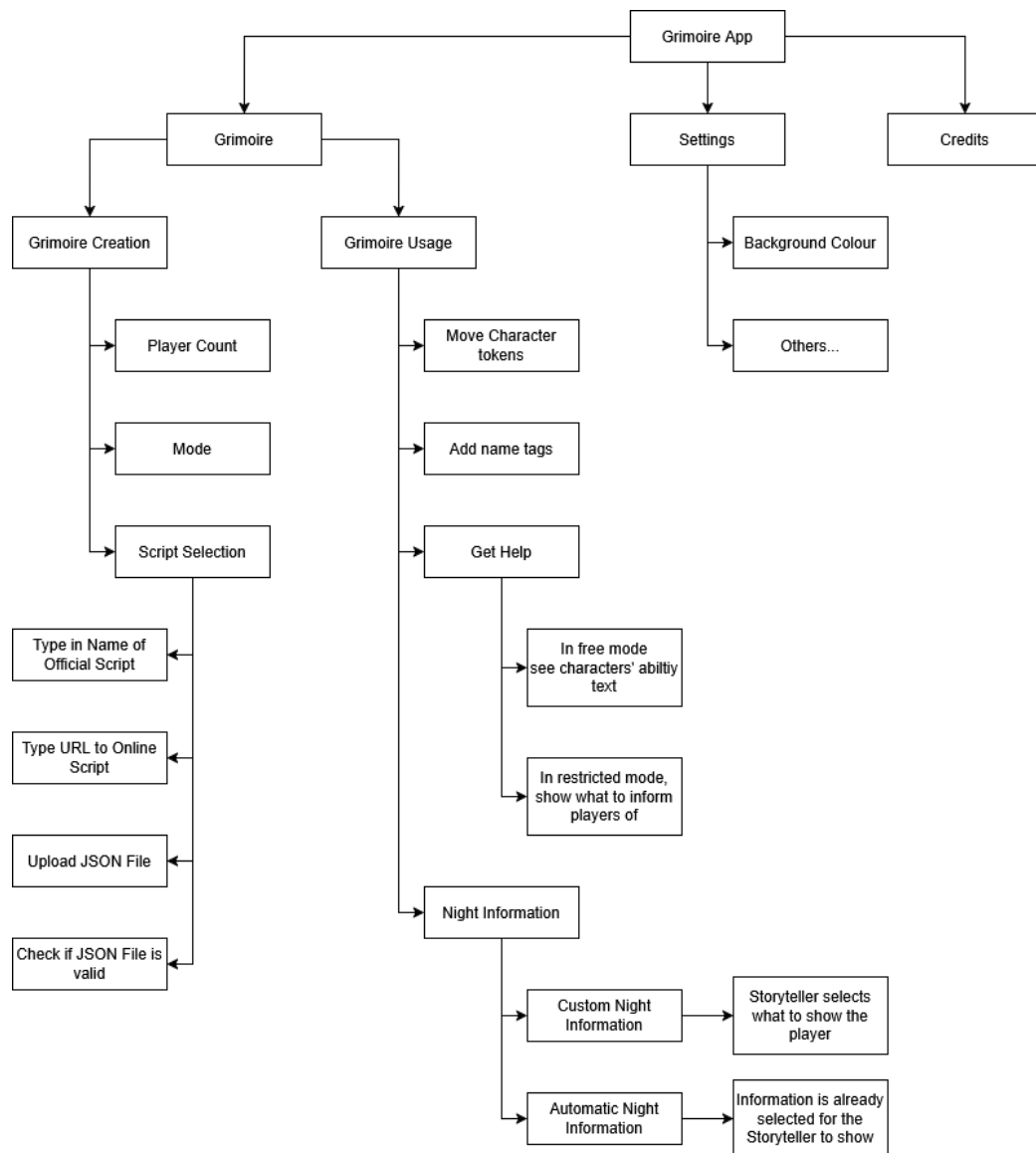
<https://release.botc.app/resources/>

Copyright and Usage Restrictions

Due to the usage restrictions of TPI's content (Paragraph 2 of Section 2 "Original Visual Assets" of [Blood on the Clocktower's Terms of Use](#)), I will be making use of Tomozbot's botc-icons on [GitHub](#) throughout.

I will also be making use of the Papyrus and Old English Text MT fonts as provided by [online-fonts.com](#).

Systems Diagram



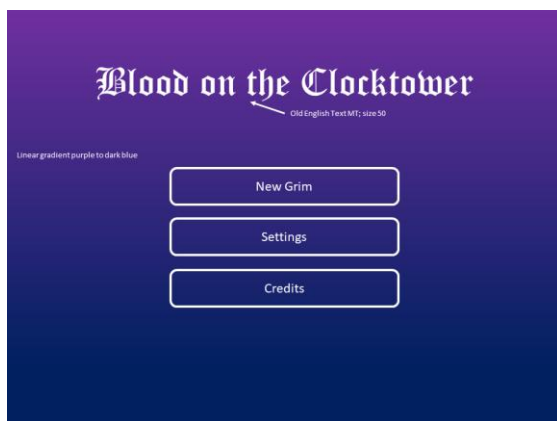
User Interface

My main inspiration for the user interface will be the physical grimoire (pictured in [Appendix 1.C](#)) that comes with Blood on the Clocktower itself. This means that the design will be quite minimal with unintrusive menu buttons and pop-up menus for most actions that could be performed.

PowerPoint Mock-Up

In order to assist later stages, I made a rough mock-up in PowerPoint, which includes some possible user actions and their result

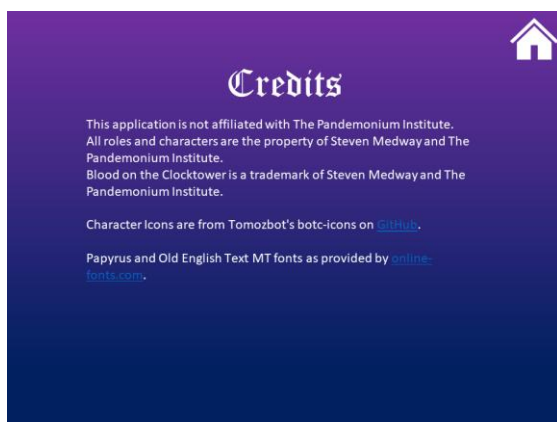
Home



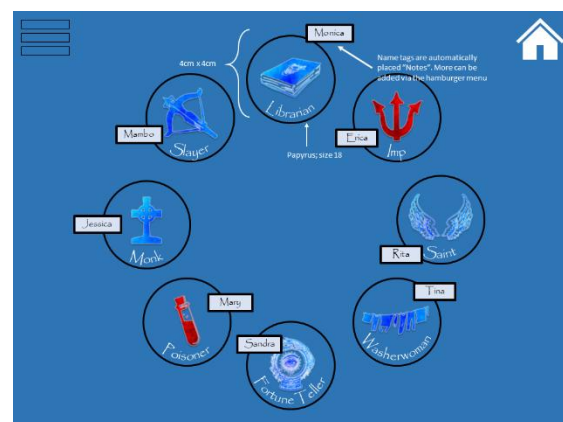
Character Selection



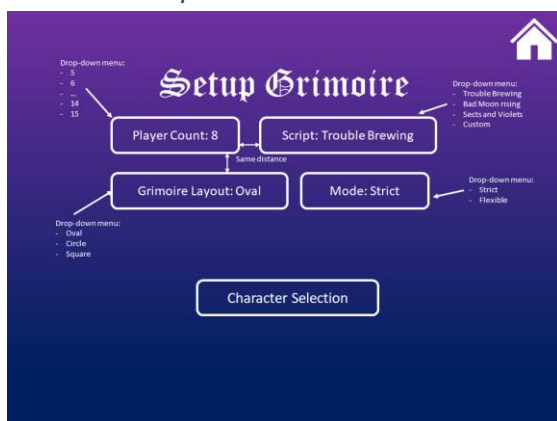
Credits



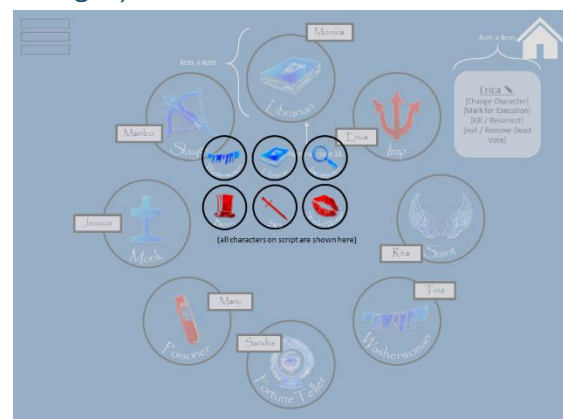
Grimoire



Grimoire Setup



Grimoire (with "change character" dialogue)



Grimoire Page Design

The grimoire page is where all the players' character tokens will be held and where the user will spend most of their time within the app.

Character Token Layout

In order for the character tokens to be placed in the shapes required for Objective 1.4, I will be using the parametric equations for both a circle and an ellipse as shown (where r_c , r_x and r_y are the radii of the circle and the ellipse).

Circle:

$$\begin{aligned} x(t) &= r_c \sin(t) \\ y(t) &= r_c \cos(t), \quad \text{where } 0 \leq t \leq 2\pi \end{aligned}$$

Ellipse:

$$\begin{aligned} x(t) &= r_x \sin(t) \\ y(t) &= r_y \cos(t), \quad \text{where } 0 \leq t \leq 2\pi \end{aligned}$$

To make use of these, the tokens will be placed with a unique index value (i). These indices can then be used to obtain a value for t using $t = \frac{i}{i_{max}} * 2\pi$, placing them in their correct spot.

The values for r_c , r_x and r_y will change based on the screen size of the device.

Modes

During the creation of a Grimoire, the user will be given the option to be in either the Strict or Flexible mode. The Strict mode ensures that tokens cannot be moved around and must stay in their specific area. Whereas the Flexible mode only initially places the tokens in the chosen shape but allows the user to freely move them about afterwards.

In addition to the character tokens, the user has the option to add reminder tokens to help them. In Strict mode they will be tied to the character token they belong to pointing towards the centre of the grimoire.

Technical Solution

This section will demonstrate the programming tasks that have been completed based on the design and analysis stages to ensure all objectives have been completed to their fullest extents.

Code Evidence WITH ANNOTATIONS and links to appendix
Iterative Testing

Home Page

...

Dialogues

Settings

New Grimoire Setup

Grimoire

...

Dialogues

Modify Player

Add Token

Wake Player

Final Testing

Test Title	Object being Tested	Aspect being Tested	Expected Outcome	Actual Outcome	Pass / Fail	Image
						Appendix 4.A
						Appendix 4.B

Evaluation

Evaluation against success criteria

Effectiveness of Solution

Maintainability

Future Improvements

Appendix

1. Additional Images

A. Trouble Brewing

[Back to Background Description](#)

trouble brewing

TOWNSFOLK

	Washerwoman You start knowing that 1 of 2 players is a particular Townsfolk.		Monk Each night*, choose a player (not yourself): they are safe from the Demon tonight.
	Librarian You start knowing that 1 of 2 players is a particular Outsider. (Or that zero are in play.)		Ravenkeeper If you die at night, you are woken to choose a player: you learn their character.
	Investigator You start knowing that 1 of 2 players is a particular Minion.		Virgin The 1st time you are nominated, if the nominator is a Townsfolk, they are executed immediately.
	Chef You start knowing how many pairs of evil players there are.		Slayer Once per game, during the day, publicly choose a player: if they are the Demon, they die.
	Empath Each night, you learn how many of your 2 alive neighbors are evil.		Soldier You are safe from the Demon.
	Fortune Teller Each night, choose 2 players: you learn if either is a Demon. There is a good player that registers as a Demon to you.		Mayor If only 3 players live & no execution occurs, your team wins. If you die at night, another player might die instead.
	Undertaker Each night*, you learn which character died by execution today.		

OUTSIDERS

	Butler Each night, choose a player (not yourself): tomorrow, you may only vote if they are voting too.		Recluse You might register as evil & as a Minion or Demon, even if dead.
	Drunk You do not know you are the Drunk. You think you are a Townsfolk character, but you are not.		Saint If you die by execution, your team loses.

MINIONS

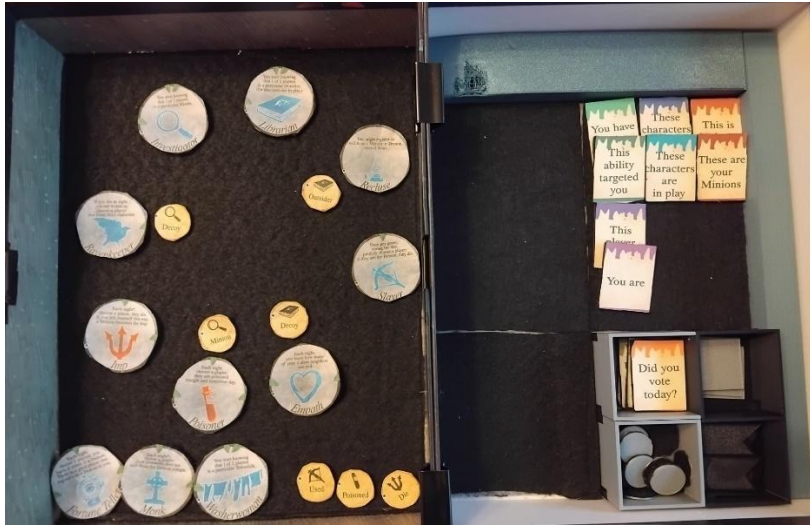
	Poisoner Each night, choose a player: they are poisoned tonight and tomorrow day.		Scarlet Woman If there are 5 or more players alive & the Demon dies, you become the Demon. (Travellers don't count.)
	Spy Each night, you see the Grimoire. You might register as good & as a Townsfolk or Outsider, even if dead.		Baron There are extra Outsiders in play. [+2 Outsiders]

DEMONS

	Imp Each night*, choose a player: they die. If you kill yourself this way, a Minion becomes the Imp.
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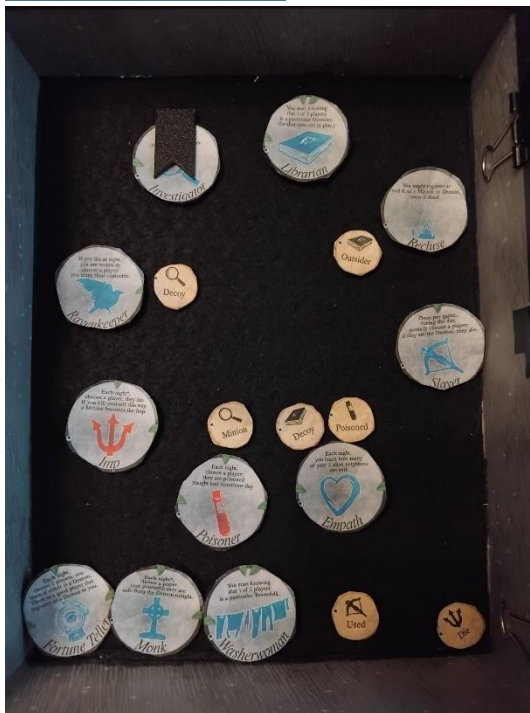
B. Physical Grimoire Full

[Back to Introduction](#)



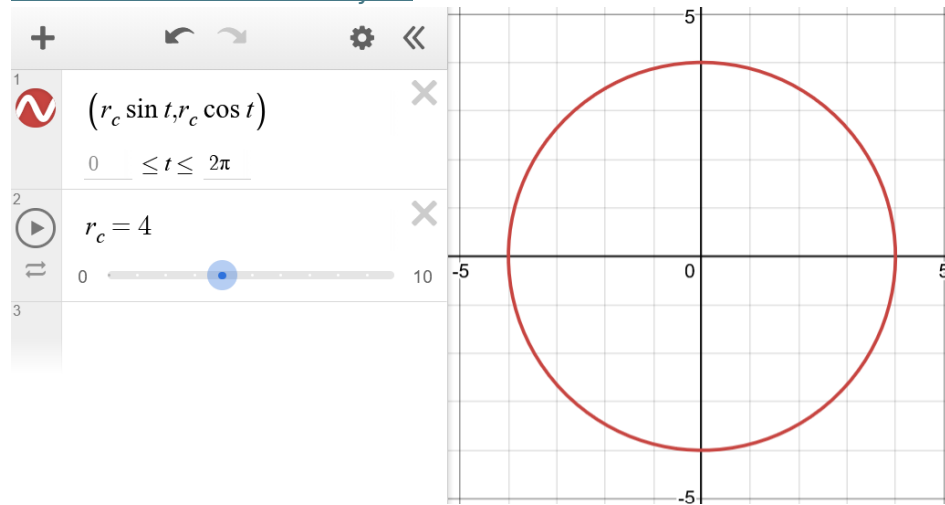
C. Physical Grimoire

[Back to User Interface](#)



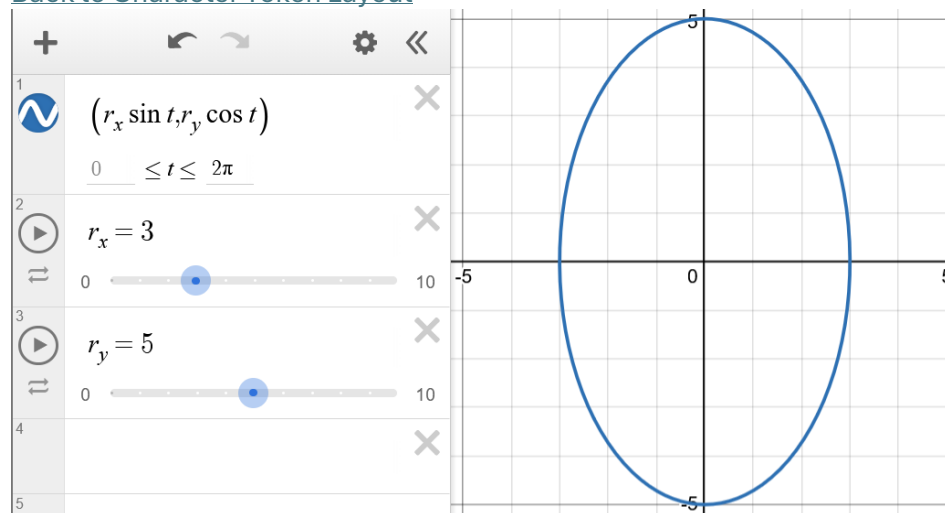
D. Circle Equation

[Back to Character Token Layout](#)



E. Ellipse Equation

[Back to Character Token Layout](#)



2. Code

3. Full Interview Transcript

Q: What assistance (if any) would be useful in storytelling (i.e. night order, what to do when waking people)?

A: [Night order and character description so you know what you need to do. Also being able to use multiple of the same reminder tokens for characters such as the Cannibal⁴](#)

Q: Would alternate colour schemes (such as light/dark mode) be useful?

A: [Yes](#)

Q: Should the app have different modes, i.e. strict vs flexible (in reference to freedom of token movement)?

A: [Yes](#)

Q: Would automatic character assignation be a good feature?

A: [Yes, as you don't need the physical tokens or need to tell \[the players\] their roles](#)

Q: How closely should the app replicate in-person usage?

A: [As closely as possible](#)

Q: (Following on from that), which one of physical or digital night information cards be preferable?

A: [Digital](#)

Q: Would having capacity for homebrew characters be useful?

A: [Yes](#)

Q: Would you rather an in-app tutorial or a document-based guide?

A: [In-app](#)

Q: Should there be automatic waking determiners (i.e. tells you what to tell the player based on their ability)

A: [Yes, but it should be optional](#)

Q: Any other suggestions or ideas for this app?

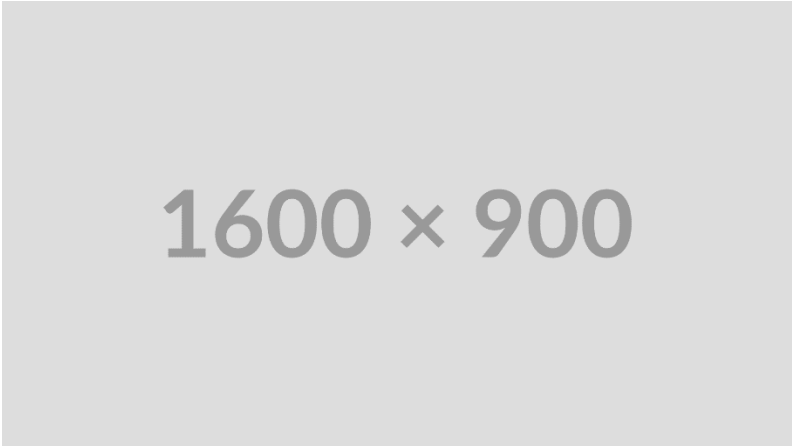
A: [Naming of tokens to relate them to the players](#)

⁴ Certain character's abilities (such as the [Cannibal](#)) use other characters abilities so may need additional copies of those characters' tokens

4. Testing

[Back to Final Testing](#)

A. Login Test Result



1600 × 900

B. Other Test Result



1500 × 800

5. References

There are no sources in the current document.